



Ch12 Pricing Decisions and Cost Management Test

Financial Accounting (National University of Computer and Emerging Sciences)



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CHAPTER 12: PRICING DECISIONS AND COST MANAGEMENT

TRUE/FALSE

1. Companies must always examine pricing decisions through the eyes of their customers.

Answer: True

2. Relevant costs for pricing decisions include manufacturing costs, but not costs from other value-chain functions.

Answer: False

Relevant costs for pricing decisions include costs from all value-chain functions, from R&D to customer service.

3. Cost information only helps the company decide how many units to produce.

Answer: True

4. In markets with little or no competition, the key factor affecting price is costs, not customers' willingness to pay or competitors.

Answer: False

In markets with little or no competition, the key factor affecting price is the customers' willingness to pay, not costs or competitors.

5. When prices are set in a competitive marketplace, product costs are the most important influence on pricing decisions.

Answer: False

When prices are set in a competitive marketplace, companies have no control over setting prices and must accept the price determined by the market.

6. Short-run pricing decisions include adjusting product mix in a competitive environment.

Answer: True

7. Profit margins are often set to earn a reasonable return on investment for short-term pricing decisions, but not long-term pricing decisions.

Answer: False

Profit margins are often set to earn a reasonable return on investment for long-term pricing decisions, but not short-term pricing decision.

8. In a one-time-only special order, existing fixed manufacturing costs are irrelevant.

Answer: True

9. Relevant costs of a bidding decision should exclude revenues lost on lower-priced sales to existing customers.

Answer: False

Relevant costs of a bidding decision should *include* revenues lost on lower-priced sales to existing customers.

10. Customers prefer stable and predictable prices over a long time horizon.

Answer: True

11. Product cost analysis is important even if market forces set prices.

Answer: True

True, because a company still has to decide how much of that product to supply to market to maximize operating income, and that decision is based on cost factors.

12. Target pricing is a form of cost-based pricing.

Answer: False

Target pricing is a form of *market-based pricing*.

13. The first step in target pricing is to determine the target cost of the product.

Answer: False

The first step in target pricing is to determine the *target price* of the product.

14. Value engineering has the objective of reducing costs while still satisfying customer needs.

Answer: True

15. Rework is an example of a value-added cost.

Answer: False

Rework is an example of a *nonvalue-added cost*.

16. It is always clear which activities add value and which do not add value to a product.

Answer: False

Activities do not always fall neatly into value-added or nonvalue-added categories.

17. Value engineering seeks to reduce value-added costs as well as nonvalue-added costs.

Answer: True

True, value-added costs can be reduced through greater efficiencies.

18. Suppliers play a key role in the success of target costing.

Answer: True

19. Costs may be locked in before they are incurred.

Answer: True

20. Locked-in costs have already been incurred.

Answer: False

Locked-in costs are those costs that have not yet been incurred, but which, based on decisions that have already been made, will be incurred in the future.

21. For manufacturing firms, product costs are generally locked in during the manufacturing stage.

Answer: False

For manufacturing firms, product costs are generally locked in during the *design* stage.

22. One goal of target costing is to design costs out of products.

Answer: True

23. Spending more on the design phase of a new product usually reduces subsequent product-related costs.

Answer: True

24. Kaizen costing focuses on improving productivity and eliminating waste through continuous improvements.

Answer: True

25. In cost-plus pricing, the markup is a rigid number that determines the actual selling price.

Answer: False

In cost-plus pricing, the markup is ultimately determined by the market.

26. The target rate of return on investment is another way of referring to the markup percentage.

Answer: False

The target rate of return on investment and the markup percentage are two different things.

27. Cost bases that include fewer costs also have lower markups.

Answer: False

Cost bases that include fewer costs have higher markups.

28. Markups tend to be lower in more competitive markets.

Answer: True

29. The full-cost formula for pricing is relatively simple to use because it does not require a detailed analysis of cost behavior.

Answer: True

30. A full-cost base rather than a variable-cost base is a better guide for discounting decisions that may affect long-term customers.

Answer: True

31. To be profitable, a company must generate revenues to cover costs incurred in all six business functions.

Answer: True

32. Life-cycle budgeting is particularly important when nonproduction costs are significant.

Answer: True

33. Many companies use life-cycle budgeting to determine target prices.

Answer: True

34. Customer life-cycle costs focus on total costs incurred by the customer from purchase to disposal.

Answer: True

35. When price discrimination is effective, cost is not a major factor in setting prices.

Answer: True

36. When demand is elastic, an increase in price will lead to an increase in profits.

Answer: False

When demand is *inelastic*, an increase in price will usually lead to an increase in profits.

37. Price discrimination is the illegal practice of charging some customers a higher price than is charged to other customers.

Answer: False

Price discrimination is a *legal* practice of charging some customers a higher price than is charged to other customers.

38. When demand is strong, firms usually increase markups.

Answer: True

True, when capacity is limited this is referred to as peak-load pricing.

39. Price discrimination laws apply only to manufacturers.

Answer: True

40. Price discrimination is only illegal if the intent is to destroy competition.

Answer: True

41. A business that engages in predatory pricing violates various U.S. antitrust laws.

Answer: True

42. Price dumping occurs when a domestic company is trying to get rid of out-of-style products at a substantially reduced price.

Answer: False

Price dumping occurs when a non-U.S. company sells a product in the United States at a price below the market value where it is produced and this action threatens to injure an industry in the United States.

43. Collusive pricing occurs when companies in an industry conspire in their pricing and output decisions to achieve a price above the competitive price.

Answer: True

MULTIPLE CHOICE

44. Companies should ONLY produce and sell units as long as
- there is customer demand for the product.
 - the competition allows it.
 - the revenue from an additional unit exceeds the cost of producing it.
 - there is a generous supply of low-cost direct materials.

Answer: c

45. Too high a price may
- deter a customer from purchasing a product.
 - increase demand for the product.
 - indicate supply is too plentiful.
 - decrease a competitor's market share.

Answer: a

46. Companies must ALWAYS examine pricing
- based on the supply of the product.
 - based on the cost of producing the product.
 - through the eyes of their customers.
 - through the eyes of their competitors.

Answer: c

47. Competitors
- with alternative products can force a company to lower its prices.
 - can gain a competitive pricing advantage with knowledge of your costs and operating policies.
 - may span international borders.
 - may be all of the above.

Answer: d

48. Fluctuations in exchange rates between different currencies can influence
- the cost of products using foreign suppliers.
 - the pricing of alternative products offered by foreign competitors.
 - the demand for products of foreign competitors.
 - all of the above.

Answer: d

49. The cost of producing a product
- is an important influence on pricing.
 - affects the willingness of a company to supply a product.
 - for pricing decisions includes manufacturing costs, but not product design costs.
 - in highly competitive markets controls pricing.

Answer: b

50. In a noncompetitive environment, the key factor affecting pricing decisions is
- a. the customer's willingness to pay.
 - b. the price charged for alternative products.
 - c. the cost of producing and delivering the product.
 - d. all of the above.

Answer: a

51. In a competitive market with differentiated products like cameras, the key factor(s) affecting pricing decisions is/are
- a. the customer's willingness to pay.
 - b. the price charged for alternative products.
 - c. the cost of producing and delivering the product.
 - d. all of the above.

Answer: d

52. Three major influences on pricing decisions are
- a. competition, costs, and customers.
 - b. competition, demand, and production efficiency.
 - c. continuous improvement, customer satisfaction, and supply.
 - d. variable costs, fixed costs, and mixed costs.

Answer: a

53. Long-run pricing decisions
- a. have a time horizon of less than one year.
 - b. include adjusting product mix in a competitive environment.
 - c. and short-run pricing decisions generally have the same relevant costs.
 - d. use prices that include a reasonable return on investment.

Answer: d

54. Short-term pricing decisions
- a. use costs that may be irrelevant for long-term pricing decisions.
 - b. are more opportunistic.
 - c. tend to decrease prices when demand is strong.
 - d. have a time horizon of more than one year.

Answer: b

55. Relevant costs for pricing a special order include
- a. existing fixed manufacturing overhead.
 - b. nonmanufacturing costs that will not change even if the special order is accepted.
 - c. additional setup costs for the special order.
 - d. all of the above costs.

Answer: c

56. Which of the following factors should NOT be considered when pricing a special order?
- a. The likely bids of competitors
 - b. The incremental cost of one unit of product
 - c. Revenues that will be lost on existing sales if prices are lowered
 - d. Stable pricing to earn the desired long-run return

Answer: d

57. Long-run pricing
- a. needs to cover only incremental costs.
 - b. only utilizes the market-based approach to pricing and not the cost-based approach.
 - c. is a strategic decision.
 - d. strives for flexible pricing that can respond to temporary changes in demand.

Answer: c

58. For long-run pricing decisions, using stable prices has the advantage of
- a. minimizing the need to monitor competitors' prices frequently.
 - b. reducing the need to change cost structures frequently.
 - c. reducing competition.
 - d. helping build buyer-seller relationships.

Answer: d

59. A price-bidding decision for a one-time-only special order includes an analysis of
- a. all manufacturing costs.
 - b. all cost drivers related to the product.
 - c. all direct and indirect variable costs of each function in the value chain.
 - d. all fixed manufacturing costs.

Answer: c

60. For pricing decisions, full product costs
- a. include all costs that are traceable to the product.
 - b. include all manufacturing and selling costs.
 - c. include all direct costs plus an appropriate allocation of the indirect costs of all business functions.
 - d. allow for the highest possible product prices.

Answer: c

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 61 AND 62.

Northwoods manufactures rustic furniture. The cost accounting system estimates manufacturing costs to be \$120 per table, consisting of 60% variable costs and 40% fixed costs. The company has surplus capacity available. It is Northwoods' policy to add a 50% markup to full costs.

61. Northwoods is invited to bid on a one-time-only special order to supply 200 rustic tables. What is the lowest price Northwoods should bid on this special order?
- a. \$21,600
 - b. \$7,200
 - c. \$12,000
 - d. \$14,400

Answer: d

$$\$120 \times 60\% \times 200 \text{ tables} = \$14,400$$

62. A large hotel chain is currently expanding and has decided to decorate all new hotels using the rustic style. Northwoods is invited to submit a bid to the hotel chain. What per unit price will Northwoods MOST likely bid on this long-term order?
- a. \$72 per unit
 - b. \$108 per unit
 - c. \$180 per unit
 - d. \$120 per unit

Answer: c

$$\$120 + (\$120 \times 50\%) = \$180$$

Objectives: 2, 5

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 63 THROUGH 65.

Rogers' Heaters is approached by Ms. Yukki, a new customer, to fulfill a large one-time-only special order for a product similar to one offered to regular customers. Rogers' Heaters has excess capacity. The following per unit data apply for sales to regular customers:

Direct materials	\$200
Direct manufacturing labor	60
Variable manufacturing support	30
Fixed manufacturing support	<u>100</u>
Total manufacturing costs	390
Markup (30%)	<u>117</u>
Estimated selling price	<u>\$507</u>

63. For Rogers' Heaters, what is the minimum acceptable price of this one-time-only special order?
- \$290
 - \$390
 - \$260
 - \$507

Answer: a

$$\$200 + \$60 + \$30 = \$290$$

64. Before accepting this one-time-only special order, Rogers' Heaters should consider the impact on
- current plant capacity.
 - long-term customers.
 - competitors.
 - all of the above.

Answer: d

65. If Ms. Yukki wanted a long-term commitment for supplying this product, what price would MOST likely be quoted?
- \$290
 - \$390
 - \$260
 - \$507

Answer: d , 5

The estimated selling price of \$507.

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 66 THROUGH 68.
 Welch Manufacturing is approached by a European customer to fulfill a one-time-only special order for a product similar to one offered to domestic customers. Welch Manufacturing has a policy of adding a 10% markup to full costs and currently has excess capacity. The following per unit data apply for sales to regular customers:

<i>Variable costs:</i>	
Direct materials	\$30
Direct labor	10
Manufacturing overhead	15
Marketing costs	5
<i>Fixed costs:</i>	
Manufacturing overhead	100
Marketing costs	<u>20</u>
Total costs	180
Markup (10%)	<u>18</u>
Estimated selling price	<u>\$198</u>

66. For Welch Manufacturing, what is the minimum acceptable price of this one-time-only special order?
- \$40
 - \$55
 - \$60
 - \$66

Answer: c

$$\$30 + \$10 + \$15 + \$5 = \$60$$

67. What is the full cost of the product per unit?
- \$60
 - \$180
 - \$198
 - \$66

Answer: b

$$\$30 + \$10 + \$15 + \$5 + 100 + 20 = \$180$$

68. If the European customer wanted a long-term commitment for supplying this product, what price would MOST likely be quoted?
- \$66.00
 - \$180.00
 - \$198.00
 - \$217.80

Answer: c

The estimated selling price of \$198.

, 5

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 69 AND 70.

Berryman Products manufactures coffee tables. Berryman Products has a policy of adding a 20% markup to full costs and currently has excess capacity. The following information pertains to the company's normal operations per month:

Output units	30,000 tables
Machine-hours	8,000 hours
Direct manufacturing labor-hours	10,000 hours
Direct materials per unit	\$50
Direct manufacturing labor per hour	\$6
Variable manufacturing overhead costs	\$161,250
Fixed manufacturing overhead costs	\$600,000
Product and process design costs	\$450,000
Marketing and distribution costs	\$562,500

69. Berryman Products is approached by an overseas customer to fulfill a one-time-only special order for 2,000 units. All cost relationships remain the same except for a one-time setup charge of \$20,000. No additional design, marketing, or distribution costs will be incurred. What is the minimum acceptable bid per unit on this one-time-only special order?
- \$67.38
 - \$77.38
 - \$111.13
 - \$80.85

Answer: a

Direct materials	\$50.000
Direct manufacturing labor ($\$6 \times 10,000$) / 30,000	2.000
Variable manufacturing ($\$161,250 / 30,000$)	5.375
Setup ($\$20,000 / 2,000$)	<u>10.000</u>
Minimum acceptable bid	<u>\$67.375</u>

70. For long-run pricing of the coffee tables, what price will MOST likely be used by Berryman?
- \$67.38
 - \$80.85
 - \$111.13
 - \$133.35

Answer: d

Objectives: 2, 5

Direct materials	\$ 50.000
Direct manufacturing labor ($\$6 \times 10,000$)/30,000	2.000
Variable manufacturing ($\$161,250/30,000$)	5.375
Fixed manufacturing ($\$600,000/30,000$)	20.000
Product and process design costs ($\$450,000/30,000$)	15.000
Marketing and distribution ($\$562,500/30,000$)	<u>18.750</u>
Full cost per unit	\$111.125
Markup (20%)	<u>22.225</u>
Estimated selling price	<u>\$133.350</u>

71. Target pricing
- a. is used for short-term pricing decisions.
 - b. is one form of cost-based pricing.
 - c. estimate is based on customers' perceived value of the product.
 - d. relevant costs are all variable costs.

Answer: c

72. To understand how competitors might price competing products a company
- a. needs to understand the competitor's technologies and financial conditions.
 - b. may get information from suppliers that service the competitor.
 - c. may use reverse engineering.
 - d. may do all of the above.

Answer: d

73. The department usually in the best position to identify customers' needs is the
- a. production department.
 - b. sales and marketing department.
 - c. design department.
 - d. distribution department.

Answer: b

74. Relevant costs for target pricing are
- a. variable manufacturing costs.
 - b. variable manufacturing and variable nonmanufacturing costs.
 - c. all fixed costs.
 - d. all future costs, both variable and fixed.

Answer: d

75. Place the following steps for the implementation of target costing in order:

A = Derive a target cost
B = Develop a target price
C = Perform value engineering
D = Determine target operating income

- a. B D A C
- b. B A D C
- c. A D B C
- d. A B C D

Answer: a

76. Value engineering may result in all of the following EXCEPT
- a. improved product design.
 - b. changes in materials specifications.
 - c. increases in the quantity of nonvalue-added cost drivers.
 - d. the evaluation of all business functions within the value chain.

Answer: c

77. Value-added costs
- a. are costs that a customer is unwilling to pay for.
 - b. include maintenance and repairs of the manufacturing equipment.
 - c. are reduced through improved efficiencies.
 - d. if eliminated increase profitability.

Answer: c

78. To design costs out of products is a goal of
- a. cost-plus pricing.
 - b. target costing.
 - c. kaizen costing.
 - d. peak-load costing.

Answer: b

79. All of the following are true regarding target costing EXCEPT
- a. improvements are implemented in small incremental amounts.
 - b. customer input is essential to the target costing process.
 - c. input is requested from suppliers and distributors.
 - d. a key goal is to minimize costs over the product's useful life.

Answer: a

80. All of the following are associated with target costing EXCEPT
- a. value engineering.
 - b. the markup component.
 - c. all value-chain business functions.
 - d. cross-functional teams.

Answer: b

81. When target costing and target pricing are used together,
- a. the target cost is established first, then the target price.
 - b. the target cost is the estimated long-run cost that enables a product or service to achieve a desired profit.
 - c. the focus of target pricing is to undercut the competition.
 - d. target costs are generally higher than current costs.

Answer: b

82. The product strategy in which companies first determine the price at which they can sell a new product and then design a product that can be produced at a low enough cost to provide adequate operating income is referred to as
- a. cost-plus pricing.
 - b. target costing.
 - c. kaizen costing.
 - a. full costing.

Answer: b

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 83 THROUGH 86.
After conducting a market research study, Schultz Manufacturing decided to produce a new interior door to complement its exterior door line. It is estimated that the new interior door can be sold at a target price of \$60. The annual target sales volume for interior doors is 20,000. Schultz has target operating income of 20% of sales.

83. What are target sales revenues?
- a. \$960,000
 - b. \$2,000,000
 - c. \$1,200,000
 - d. none of the above

Answer: c

$$\$60 \times 20,000 = \$1,200,000$$

84. What is the target operating income?
- a. \$240,000
 - b. \$300,000
 - c. \$192,000
 - d. \$180,000

Answer: a

$$\$1,200,000 \times 20\% = \$240,000$$

85. What is the target cost?
- a. \$900,000
 - b. \$960,000
 - c. \$1,260,000
 - d. \$1,008,000

Answer: b

$$\$1,200,000 - \$240,000 = \$960,000$$

86. What is the target cost for each interior door?
- a. \$48
 - b. \$58
 - c. \$60
 - d. \$45

Answer: a

$$\$960,000 / 20,000 = \$48$$

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 87 THROUGH 89.
 Sheltar's TV currently sells small televisions for \$180. It has costs of \$140. A competitor is bringing a new small television to market that will sell for \$150. Management believes it must lower the price to \$150 to compete in the market for small televisions. Marketing believes that the new price will cause sales to increase by 10%, even with a new competitor in the market. Sheltar's sales are currently 100,000 televisions per year.

87. What is the target cost if target operating income is 25% of sales?
- \$37.50
 - \$45.00
 - \$112.50
 - \$135.00

Answer: c

$$\$150 - \$150(0.25) = \$112.50$$

88. What is the change in operating income if marketing is correct and only the sales price is changed?
- \$1,100,000
 - \$300,000
 - \$(1,100,000)
 - \$(2,900,000)

Answer: d

$$[100,000 \times (\$180 - \$140)] - [110,000 \times (\$150 - \$140)] = \$(2,900,000)$$

89. What is the target cost if the company wants to maintain its same income level, and marketing is correct?
- \$112.50
 - \$113.64
 - \$123.34
 - \$140.00

Answer: b

$$\text{Current income} = 100,000 \times (\$180 - \$140) = \$4,000,000$$

$$\text{Target cost } y: \$4,000,000 = (110,000 \times \$150) - 110,000y$$

$$y = \$12,500,000 / 110,000 = \$113.64$$

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 90 THROUGH 92.
 Frank's Computer Monitors, Inc., currently sells 17" monitors for \$270. It has costs of \$210. A competitor is bringing a new 17" monitor to market that will sell for \$225. Management believes it must lower the price to \$225 to compete in the market for 17" monitors. Marketing believes that the new price will cause sales to increase by 10%, even with a new competitor in the market. Frank's Computer Monitor, Inc.'s sales are currently 10,000 monitors per year.

90. What is the target cost if operating income is 25% of sales?
- \$56.25
 - \$67.50
 - \$168.75
 - \$202.50

Answer: c
 $\$225 - \$225(0.25) = \$168.75$

91. What is the change in operating income if marketing is correct and only the sales price is changed?
- \$165,000
 - \$45,000
 - \$(165,000)
 - \$(435,000)

Answer: d
 $[10,000 \times (\$270 - \$210)] - [11,000 \times (\$225 - \$210)] = (\$435,000)$

92. What is the target cost if the company wants to maintain its same income level, and marketing is correct (rounded to the nearest cent)?
- \$168.75
 - \$170.45
 - \$185.00
 - \$210.00

Answer: b
 Current income = $10,000 \times (\$270 - \$210) = \$600,000$
 Target cost y: $\$600,000 = (11,000 \times \$225) - 11,000y$
 $y = \$1,875,000/11,000 = \170.4545

93. Concerns about target costing include all EXCEPT
- cross-functional teams may add too many features.
 - excessive pressure is put on suppliers.
 - development time may decrease.
 - burnout of design engineers.

Answer: c

94. Direct material costs are locked in when they are
- a. designed.
 - b. assembled.
 - c. sold.
 - d. delivered.

Answer: a

95. Cost accounting systems focus on when costs
- a. are incurred.
 - b. are locked in.
 - c. are paid for.
 - d. are used for setting prices for products and services.

Answer: a

96. Most of a product's life-cycle costs are locked in by decisions made during the _____ business function of the value chain.
- a. design
 - b. manufacturing
 - c. customer-service
 - d. marketing

Answer: a

97. For most products, the majority of costs are incurred during the _____ business function of the value chain.
- a. design
 - b. manufacturing
 - c. customer-service
 - d. marketing

Answer: b

98. _____ focuses on reducing costs during the manufacturing stage.
- a. Target costing
 - b. Kaizen costing
 - c. Cost-plus pricing
 - d. Life-cycle costing

Answer: b

99. Cross-functional engineering teams may include
- a. marketing managers.
 - b. suppliers.
 - c. management accountants.
 - d. all of the above.

Answer: d

100. In some industries such as legal and consulting, most costs are locked in
- a. when they are incurred.
 - b. during the design stage.
 - c. during the customer-service stage.
 - d. during the marketing stage.

Answer: a

101. Value engineering can reduce all EXCEPT
- a. existing fixed manufacturing costs.
 - b. value-added costs.
 - c. nonvalue-added costs.
 - d. rework-hours.

Answer: a

102. A graph comparing locked-in costs with incurred costs will have
- a. locked-in costs rising much faster initially, but dropping to zero after the product is manufactured.
 - b. the two cost lines running parallel until the end of the process, when they join.
 - c. locked-in costs rising much faster initially than the incurred cost, but joining the incurred cost line at the completion of the value-chain functions.
 - d. no differences unless the product is manufactured inefficiently.

Answer: c

103. Graphic analysis of incurred and locked-in costs provides several insights as to how the different concepts influence decisions. Which of the following statements is FALSE?
- a. Costs are generally locked in before they are incurred.
 - b. After a product's design has been approved, costs are difficult to influence.
 - c. When and how costs are locked in are more important than when and how costs are incurred.
 - d. Most costs are locked in during the manufacturing process.

Answer: d

104. Value engineering can reduce costs by all EXCEPT
- a. simplifying the design and thereby decreasing the number of component parts.
 - b. reducing the number of features offered.
 - c. redesigning alternative options over and over until the wishes of all cross-functional team members are accommodated.
 - d. building efficiencies into value-added costs.

Answer: c

105. The cost-plus pricing approach is generally in the form:
- a. $\text{Cost base} + \text{Markup component} = \text{Prospective selling price}.$
 - b. $\text{Prospective selling price} - \text{Cost base} = \text{Markup component}.$
 - c. $\text{Cost base} + \text{Gross margin} = \text{Prospective selling price}.$
 - d. $\text{Variable cost} + \text{Fixed cost} + \text{Contribution margin} = \text{Prospective selling price}.$

Answer: a

106. In cost-plus pricing, the markup component
- a. is a rigid number.
 - b. is ultimately determined by the market.
 - c. provides a means to calculate the actual selling price.
 - d. is the end rather than the start of pricing decisions.

Answer: b

107. A product's markup percentage needs to cover nonmanufacturing variable costs when the cost base is
- a. the full cost of the product.
 - b. the variable cost of the product.
 - c. variable manufacturing costs.
 - d. any of the above cost bases.

Answer: c

108. A product's markup percentage needs to cover operating profits when the cost base is
- a. the full cost of the product.
 - b. the variable cost of the product.
 - c. variable manufacturing costs.
 - d. any of the above cost bases.

Answer: d

109. Erickson Company is considering pricing its 5,000-gallon petroleum tanks using either variable manufacturing or full product costs as the base. The variable cost base provides a prospective price of \$3,000 and the full cost base provides a prospective price of \$3,050. The difference between the two prices is
- a. the estimated amount of profit.
 - b. that the variable cost base estimates fixed costs in the markup percentage while the full cost base includes an amount for fixed costs.
 - c. known as price discrimination.
 - d. caused by the inability of most companies to estimate fixed cost per unit with any degree of reliability.

Answer: b

110. _____ starts with estimated product costs and next adds desired operating income.
- a. Cost-plus pricing
 - b. Target costing
 - c. Kaizen costing
 - d. Life-cycle budgeting

Answer: a

111. The amount of markup percentage is usually higher if
- a. there is idle capacity.
 - b. demand is strong.
 - c. competition is intense.
 - d. demand is elastic.

Answer: b

112. The markup percentage is usually higher if the cost base used is
- a. the full cost of the product.
 - b. the variable cost of the product.
 - c. variable manufacturing costs.
 - d. total manufacturing costs.

Answer: c

113. Which of the following statements is FALSE regarding cost-plus pricing?
- a. A company selects a cost base that it regards as reliable.
 - b. A company uses a markup percentage that estimates a product price that covers full product costs and earns the required return on investment.
 - c. The selling price computed is only a prospective price.
 - d. The cost-plus price chosen has already been studied for customer reaction to the price.

Answer: d

114. Advantages of using the full cost of the product as the cost base include all EXCEPT
- a. that managers are informed regarding the minimum long-run cost they need to recover to stay in business.
 - b. that it limits the ability of a salesperson to cut prices.
 - c. that fixed cost allocations can be arbitrary.
 - d. that it does not require a detailed analysis of cost behavior for computations.

Answer: c

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 115 THROUGH 117.
Timothy Company has invested \$2,000,000 in a plant to make vending machines. The target operating income desired from the plant is \$300,000 annually. The company plans annual sales of 1,500 vending machines at a selling price of \$2,000 each.

115. What is the target rate of return on investment for Timothy Company?
- a. 15.0%
 - b. 17.6%
 - c. 10.0%
 - d. 11.1%

Answer: a

$$\$300,000 / \$2,000,000 = 15\%$$

116. What is the markup percentage as a percentage of cost for Timothy Company?
- a. 15.0%
 - b. 17.6%
 - c. 10.0%
 - d. 11.1%

Answer: d

$$\$300,000 / [(1,500 \times \$2,000) - \$300,000] = 11.1\%$$

117. What is the cost base of each vending machine for Timothy Company?
- a. \$1,739
 - b. \$1,802
 - c. \$1,700
 - d. \$1,780

Answer: b

$$\$2,000 / 1.11 = \$1,802$$

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 118 THROUGH 120.
Grant Company has invested \$1,000,000 in a plant to make commercial juicer machines. The target operating income desired from the plant is \$180,000 annually. The company plans annual sales of 7,000 juicer machines at a selling price of \$200 each.

118. What is the target rate of return on investment for Grant Company?
- a. 22.0%
 - b. 18.0%
 - c. 14.8%
 - d. 12.9%

Answer: b

$$\$180,000 / \$1,000,000 = 18\%$$

119. What is the markup percentage as a percentage of cost for Grant Company?
- a. 22.0%
 - b. 18.0%
 - c. 14.8%
 - d. 12.9%

Answer: c

$$\$180,000 / [(7,000 \times \$200) - \$180,000] = 14.8\%$$

120. What is the cost base of each juicer machine for Grant Company?
- a. \$174
 - b. \$162
 - c. \$169
 - d. \$152

Answer: a

$$\$200 / 1.148 = \$174$$

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 121 THROUGH 124.
Meyer Corporation budgeted the following costs for the production of its one and only product for the next fiscal year:

Direct materials	\$ 562,500
Direct labor	390,000
Manufacturing overhead	
Variable	420,000
Fixed	322,500
Selling and administrative	
Variable	180,000
Fixed	<u>240,000</u>
Total costs	<u>\$2,115,000</u>

Meyer has an annual target operating income of \$450,000.

121. The markup percentage for setting prices as a percentage of *total manufacturing costs* is
- a. 51%.
 - b. 125%.
 - c. 185%.
 - d. 245%.

Answer: a

$$(\$450,000 + \$180,000 + \$240,000) /$$
$$(\$562,500 + \$390,000 + \$420,000 + \$322,500) = 51.3\%$$

122. The markup percentage for setting prices as a percentage of *variable manufacturing costs* is
- a. 54%.
 - b. 87%.
 - c. 169%.
 - d. 122%.

Answer: c

$$\frac{(\$450,000 + \$420,000 + \$322,500 + \$180,000 + \$240,000)}{(\$562,500 + \$390,000)} = 169.3\%$$

123. The markup percentage for setting prices as a percentage of the *variable cost of the product* is
- a. 328%.
 - b. 36%.
 - c. 228%.
 - d. 65%.

Answer: d

$$\frac{(\$450,000 + \$322,500 + \$240,000)}{(\$562,500 + \$390,000 + \$420,000 + \$180,000)} = 65.2\%$$

124. The markup percentage for setting prices as a percentage of the *full cost of the product* is
- a. 328%.
 - b. 36%.
 - c. 228%.
 - d. 21%.

Answer: d

$$\$450,000 / \$2,115,000 = 21.3\%$$

125. Life-cycle costing is the name given to
- a. a method of cost planning to reduce manufacturing costs to targeted levels.
 - b. the process of examining each component of a product to determine whether its cost can be reduced.
 - c. the process of managing all costs along the value chain.
 - d. a system that focuses on reducing costs during the manufacturing cycle.

Answer: c

126. An understanding of life-cycle costs can lead to
- a. additional costs during the manufacturing cycle.
 - b. less need for evaluation of the competition.
 - c. cost effective product designs that are easier to service.
 - d. mutually beneficial relationships between buyers and sellers.

Answer: c

127. Life-cycle budgeting is particularly important when
- a. the development period for R&D is short and inexpensive.
 - b. there are significant nonproduction costs.
 - c. most costs are locked in during production.
 - d. a low percentage of costs are incurred before any revenues are received.

Answer: b

128. Life-cycle budgeting and life-cycle costing help highlight
- a. an increase in customer-service costs due to using inferior materials.
 - b. high production costs caused by a complex design.
 - c. large ordering costs due to the great number of component parts used.
 - d. an increase in annual operating income resulting from the new product.

Answer: d

129. Life-cycle budgeting
- a. has little in common with target pricing.
 - b. is most useful to companies that manufacture small items such as household plastics.
 - c. helps companies estimate revenues over a multiyear horizon.
 - d. gives companies more insight into total costs when manufacturing costs consume the majority of the resources.

Answer: c

130. Customer life-cycle costs
- a. are the costs incurred by the selling company to satisfy the customer.
 - b. are the costs to the customer for buying and using a product.
 - c. are the same as the selling life-cycle prices.
 - d. are the replacement costs of using a product or service.

Answer: b

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 131 THROUGH 133.

Bicker, Inc., is in the process of evaluating a new product using the following information.

- A new transformer has two production runs each year, each with \$10,000 in setup costs.
- The new transformer incurred \$30,000 in development costs and is expected to be produced over the next three years.
- Direct costs of producing the transformers are \$40,000 per run of 5,000 transformers each.
- Indirect manufacturing costs charged to each run are \$45,000.
- Destination charges for each transformer average \$1.00.
- Customer service expenses average \$0.20 per transformer.
- The transformers are selling for \$25 the first year and will increase by \$3 each year thereafter.
- Sales units equal production units each year.

131. What are estimated life-cycle revenues?

- a. \$250,000
- b. \$280,000
- c. \$310,000
- d. \$840,000

Answer: d

First year (5,000 x 2 runs x \$25)	\$250,000
Second year (5,000 x 2 x \$28)	280,000
Third year (5,000 x 2 x \$31)	<u>310,000</u>
Total	<u>\$840,000</u>

132. What is the estimated life-cycle operating income for the first year?

- a. \$18,000
- b. \$20,000
- c. \$48,000
- d. \$119,000

Answer: a

Sales (5,000 units x 2 runs x \$25)		\$250,000
Development costs	\$30,000	
Setup costs (2 x \$10,000)	20,000	
Direct manufacturing costs (2 x \$40,000)	80,000	
Indirect manufacturing costs (2 x \$45,000)	90,000	
Destination charges (\$1.00 x 10,000)	10,000	
Customer service (\$0.20 x 10,000)	<u>2,000</u>	<u>232,000</u>

Estimated life-cycle operating income for the first year \$ 18,000

133. What is the estimated life-cycle operating income for the first three years?

- a. \$174,000
- b. \$204,000
- c. \$636,000
- d. \$840,000

Answer: b

	<u>Year 1</u>	<u>Year 2</u>	<u>Year 3</u>	<u>Totals</u>
Life-cycle revenue	\$250,000	\$280,000	\$310,000	\$840,000
Life-cycle costs:				
Development	30,000			30,000
Setup	20,000	20,000	20,000	60,000
Direct manufacturing costs	80,000	80,000	80,000	240,000
Indirect manufacturing	90,000	90,000	90,000	270,000
Destination charges	10,000	10,000	10,000	30,000
Customer service	<u>2,000</u>	<u>2,000</u>	<u>2,000</u>	<u>6,000</u>
Total costs	<u>\$232,000</u>	<u>\$202,000</u>	<u>\$202,000</u>	<u>636,000</u>
Life-cycle operating income				<u>\$204,000</u>

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 134 THROUGH 136.

Neises, White, Granberry and Associates are in the process of evaluating its new client services for the business consulting division.

- Estate Planning, a new service, incurred \$600,000 in development costs and employee training.
- The direct costs of providing this service, which is all labor, averages \$100 per hour.
- Other costs for this service are estimated at \$2,000,000 per year.
- The current program for estate planning is expected to last for two years. At that time, a new law will be in place that will require new operating guidelines for the tax consulting.
- Customer service expenses average \$400 per client, with each job lasting an average of 400 hours. The current staff expects to bill 40,000 hours for each of the two years the program is in effect. Billing averages \$140 per hour.

134. What are estimated life-cycle revenues?

- a. \$6,400,000
- b. \$8,000,000
- c. \$11,200,000
- d. \$22,400,000

Answer: c

First year (40,000 x \$140)	\$ 5,600,000
Second year (40,000 x \$140)	<u>5,600,000</u>
Total	<u>\$11,200,000</u>

135. What is estimated life-cycle operating income for the first year?

- a. \$(1,040,000)
- b. \$(1,400,000)
- c. \$5,600,000
- d. \$6,640,000

Answer: a

Revenue (40,000 hours x \$140)		\$5,600,000
Development costs	\$ 600,000	
Direct costs (40,000 x \$100)	4,000,000	
Indirect costs	2,000,000	
Customer service (\$400 x 100 clients)	<u>40,000</u>	<u>6,640,000</u>
Operating income (loss)		<u>\$(1,040,000)</u>

136. What is the estimated life-cycle operating income for the first two years?

- a. \$(1,480,000)
- b. \$(1,400,000)
- c. \$3,200,000
- d. \$11,200,000

Answer: a

Objective: 7

	<u>Year 1</u>	<u>Year 2</u>	<u>Totals</u>
Life-cycle revenue	\$5,600,000	\$5,600,000	\$11,200,000
Life-cycle costs:			
Development	600,000		600,000
Direct costs	4,000,000	4,000,000	8,000,000
Indirect costs	2,000,000	2,000,000	4,000,000
Customer service	<u>40,000</u>	<u>40,000</u>	<u>80,000</u>
Total costs	<u>\$6,640,000</u>	<u>\$6,040,000</u>	<u>\$12,680,000</u>
Life-cycle operating income			<u>\$(1,480,000)</u>

137. Price discrimination is the practice of

- a. setting different prices for different products.
- b. charging different prices for quantity amounts.
- c. using variable costing for some products and full costing for other products when setting prices.
- d. charging different prices to different customers or clients for the same products or services.

Answer: d

138. Iowa Utility Company charges its high-usage commercial customers a lower rate per kilowatt-hour than other customers. This is an example of
- a. customer-preference pricing.
 - b. high-load pricing.
 - c. peak-load pricing.
 - d. price discrimination.

Answer: d

139. When demand for a product is inelastic and prices are increased, usually
- a. demand will increase, and operating profits will increase.
 - b. demand will remain the same, and operating profits will increase.
 - c. demand will decrease, and operating profits will decrease.
 - d. demand will remain the same, and operating profits will decrease.

Answer: b

140. When demand for a product is very elastic and prices are increased,
- a. demand will remain the same, and operating profits will increase.
 - b. demand will remain the same, and operating profits may either increase or decrease.
 - c. demand will decrease, and operating profits will decrease.
 - d. demand will decrease, and operating profits may either increase or decrease.

Answer: d

141. Costs are a major factor
- a. when demand is price-inelastic.
 - b. when demand is price-elastic.
 - c. when the opportunity for price discrimination exists.
 - d. for peak-load pricing.

Answer: b

THE FOLLOWING INFORMATION APPLIES TO QUESTIONS 142 AND 143.

LeBlanc Lighting manufactures table lamps and is considering raising the price by \$10 a unit for the coming year. With a \$10 price increase, demand is expected to fall by 2,000 units.

	<u>Currently</u>	<u>Projected</u>
Demand	20,000 units	18,000 units
Selling price	\$150	\$160
Variable costs per unit	\$100	\$100

142. Would you recommend the \$10 price increase?
- a. No, because demand decreased.
 - b. No, because the selling price increases.
 - c. Yes, because contribution margin per unit increases.
 - d. Yes, because operating income increases.

Answer: d

$[18,000 \times (\$160 - \$100)] - [20,000 \times (\$150 - \$100)] = \$80,000$ operating income increase

143. The demand for this product
- a. is greatly inelastic.
 - b. is slightly inelastic.
 - c. is elastic.
 - d. is impossible to determine.

Answer: c

144. The Maize Eagles are evaluating ticket prices for its basketball games. Studies show that Friday and Saturday night games average more than twice the fans of games on other days. The following information pertains to the stadium's normal operations per season.

Average fans per game (all games)	2,500 fans
Average fans per Friday and Saturday night games	3,500 fans
Number of home games per season	30 games
Stadium capacity	3,500 seats
Variable operating costs per operating hour	\$2,000
Marketing costs per season for basketball	\$138,750
Customer-service costs per season for basketball	\$25,000

The stadium is open for 5 operating hours on each day a game is played. All employees work by the hour except for the administrators. A maximum of one game is played per day and each fan has only one ticket per game.

The stadium authority wants to charge more for games on Friday and Saturday. What is the minimum price that should be charged for peak attendance nights?

- a. \$4.40
- b. \$8.60
- c. \$6.18
- d. \$171.45

Answer: c

Variable operating costs (30 x 5 x \$2,000)	\$300,000
Marketing	138,750
Customer service	<u>25,000</u>
<i>Total</i>	<u>\$463,750</u>

Attendance = 30 x 2,500 = 75,000 fans

Minimum price is $\$463,750 / 75,000 = \6.18

145. All are true regarding price discrimination EXCEPT that
- a. the laws apply to service providers, but not manufacturers.
 - b. it is permissible if price differences can be explained.
 - c. it is illegal only if the intent is to destroy competition.
 - d. it is most likely to occur when the cost base is the full cost of the product.

Answer: d

146. Predatory pricing is a type of price discrimination that
- a. allows prices to be cut to the level of variable costs.
 - b. is required when a company declares bankruptcy so that it can sell its remaining goods quickly.
 - c. is used in the food industry for perishable goods.
 - d. deliberately sets prices very low, sometimes even below costs, so as to minimize competition.

Answer: d

147. Hitz Video Rental is evaluating rental prices. Historical data show that Friday and Saturday have twice the rentals of other days of the week. The following information pertains to the store's normal operations per week.

Average rentals per day on Friday and Saturday	1,150
Average rentals per day on Sunday through Thursday	500
Store hours per day	12
Total units available for rent	10,000
Variable operating costs per hour	\$ 40
Marketing costs per week	\$1,500
Customer service costs per week	\$ 250

The store manager wants to charge more for rentals on Friday and Saturday. What is the minimum price that should be charged during peak rental days?

- a. \$0.60
- b. \$0.83
- c. \$0.90
- d. \$1.06

Answer: d

Variable costs (\$40 x 12 x 7)	\$3,360
Marketing	1,500
Customer service	<u>250</u>
<i>Total costs per week</i>	<u>\$5,110</u>
<i>Average rental cost per customer</i> $\$5,110 / [(2 \times 1,150) + (5 \times 500)] = \1.06	

148. To minimize the chances of violating pricing laws, a company should
- a. keep detailed records of variable costs for all value-chain business functions.
 - b. use a variable cost-plus markup method of pricing.
 - c. underprice products on a consistent basis, rather than sporadically.
 - d. use dumping only when a product is at the end of its life cycle.

Answer: a

149. Collusive pricing occurs when
- a. a company wants two products to sell for the same, or almost the same, amount.
 - b. a company wants a product to sell for the same as a competitor's product.
 - c. two or more companies agree to sell a product at a price higher than should be expected.
 - d. competitors are part of the same large parent organization.

Answer: c

EXERCISES AND PROBLEMS

150. Backwoods Incorporated manufactures rustic furniture. The cost accounting system estimates manufacturing costs to be \$80 per table, consisting of 70% variable costs and 30% fixed costs. The company has surplus capacity available. It is Backwoods' policy to add a 50% markup to full costs.
- a. Backwoods Incorporated is invited to bid on an order to supply 100 rustic tables. What is the lowest price Backwoods should bid on this one-time-only special order?
 - b. A large hotel chain is currently expanding and has decided to decorate all new hotels using the rustic style. Backwoods Incorporated is invited to submit a bid to the hotel chain. What is the lowest price per unit Backwoods should bid on this long-term order?

Answer:

- a. The lowest price Backwoods should bid on the 100 table one-time special order is \$5,600 = Variable costs ($\$80 \times .70 \times 100$ tables), the short-term incremental costs.
- b. The lowest price Backwoods should bid on the long-term hotel chain order is \$120 per table = Full costs $\$80 + 50\%$ markup, the long-term targeted price.

Objectives: 2, 3

151. Schlickau Company manufactures basketball backboards. The following information pertains to the company's normal operations per month:

Output units	15,000 boards
Machine-hours	4,000 hours
Direct manufacturing labor-hours	5,000 hours
Direct manufacturing labor per hour	\$12
Direct materials per unit	\$100
Variable manufacturing overhead costs	\$150,000
Fixed manufacturing overhead costs	\$300,000
Product and process design costs	\$200,000
Marketing and distribution costs	\$250,000

Required:

- For long-run pricing, what is the full-cost base per unit?
- Schlickau Company is approached by an overseas city to fulfill a one-time-only special order for 1,000 units. All cost relationships remain the same except for an additional one-time setup charge of \$40,000. No additional design, marketing, or distribution costs will be incurred. What is the minimum acceptable bid per unit on this one-time-only special order?

Answer:

a. Direct materials	\$150,000
Direct manufacturing labor ($\$12 \times 5,000/15,000$)	4,000
Variable manufacturing ($\$150,000/15,000$)	10,000
Fixed manufacturing ($\$300,000/15,000$)	20,000
Marketing and distribution ($\$250,000/15,000$)	16,667
Research and development ($\$200,000/15,000$)	<u>13,333</u>
Total	<u>\$214,000</u>
b. Direct materials	\$100,000
Direct manufacturing labor	4,000
Variable manufacturing	10,000
Setup	<u>40,000</u>
Total	<u>\$154,000</u>

152. Steven Corporation manufactures fishing poles that have a price of \$21.00. It has costs of \$16.32. A competitor is introducing a new fishing pole that will sell for \$18.00. Management believes it must lower the price to \$18.00 in order to compete in the highly cost-conscious fishing pole market. Marketing believes that the new price will maintain the current sales level. Steven Corporation's sales are currently 200,000 poles per year.

Required:

- What is the target cost for the new price if target operating income is 20% of sales?
- What is the change in operating income for the year if \$18.00 is the new price and costs remain the same?
- What is the target cost per unit if the selling price is reduced to \$18.00 and the company wants to maintain its same income level?

Answer:

- $\$18.00 - \$18.00(0.20) = \$14.40$
- $$\begin{aligned}\text{Change} &= 200,000 \times (\$21.00 - \$16.32) - [200,000 \times (\$18.00 - \$16.32)] \\ &= \$936,000 - \$336,000 \\ &= \$600,000 \text{ reduction in income}\end{aligned}$$
- $\text{Current income} = 200,000 \times (\$21.00 - \$16.32) = \$936,000$

Target cost per unit:

$$\begin{aligned}\$936,000 &= (200,000 \times \$18.00) - 200,000y \\ 200,000y &= \$2,664,000 \\ y &= \$13.32\end{aligned}$$

153. Robert's Medical Equipment Company manufactures hospital beds. Its most popular model, Deluxe, sells for \$5,000. It has variable costs totaling \$2,800 and fixed costs of \$1,000 per unit, based on an average production run of 5,000 units. It normally has four production runs a year, with \$400,000 in setup costs each time. Plant capacity can handle up to six runs a year for a total of 30,000 beds.

A competitor is introducing a new hospital bed similar to Deluxe that will sell for \$4,000. Management believes it must lower the price in order to compete. Marketing believes that the new price will increase sales by 25% a year. The plant manager thinks that production can increase by 25% with the same level of fixed costs. The company currently sells all the Deluxe beds it can produce.

Required:

- What is the annual operating income from Deluxe at the current price of \$5,000?
- What is the annual operating income from Deluxe if the price is reduced to \$4,000 and sales in units increase by 25%?
- What is the target cost per unit for the new price if target operating income is 20% of sales?

Answer:

- | | | |
|--------------------------------------|------------------|----------------------|
| a. Sales (20,000 x \$5,000) | | \$100,000,000 |
| Costs: | | |
| Variable costs (20,000 x \$2,800) | \$56,000,000 | |
| Fixed costs (\$1,000 x 5,000 x 4) | 20,000,000 | |
| Setup costs (\$400,000 x 4) | <u>1,600,000</u> | <u>77,600,000</u> |
| Operating income | | <u>\$ 22,400,000</u> |
| b. Sales (25,000 x \$4,000) | | \$100,000,000 |
| Costs: | | |
| Variable costs (25,000 x \$2,800) | \$70,000,000 | |
| Fixed costs, same | 20,000,000 | |
| Setup costs (\$400,000 x 5) | <u>2,000,000</u> | <u>92,000,000</u> |
| Operating income | | <u>\$ 8,000,000</u> |
| c. \$4,000 - \$4,000(0.20) = \$3,200 | | |

154. Reuter Avionics currently sells radios for \$1,800. It has costs of \$1,400. A competitor is bringing a new radio to market that will sell for \$1,600. Management believes it must lower the price to \$1,600 to compete in the market for radios. Marketing believes that the new price will cause sales to increase by 10%, even with a new competitor in the market. Reuter's sales are currently 1,000 radios per year.

Required:

- What is the target cost if target operating income is 25% of sales?
- What is the change in operating income if marketing is correct and only the sales price is changed?
- What is the target cost if the company wants to maintain its same income level, and marketing is correct?

Answer:

- $\$1,600 - \$1,600 (0.25) = \$1,200$
- $(1,000 \times (\$1,800 - \$1,400)) - (1,100 \times (\$1,600 - \$1,400)) = \$180,000$
- Current income = $1,000 \times (\$1,800 - \$1,400) = \$400,000$
Target cost y: $\$400,000 = (1,100 \times \$1,600) - 1,100y$
 $y = \$1,360,000/1,100$
 $y = \$1,236.3636$

155. Nancy Company has budgeted sales of \$300,000 with the following budgeted costs:

Direct materials	\$60,000
Direct manufacturing labor	40,000
Factory overhead	
Variable	30,000
Fixed	50,000
Selling and administrative expenses	
Variable	20,000
Fixed	30,000

Compute the average markup percentage for setting prices as a percentage of:

- The full cost of the product
- The variable cost of the product
- Variable manufacturing costs
- Total manufacturing costs

Answer:

- $\$60,000 + \$40,000 + \$30,000 + \$50,000 + \$20,000 + \$30,000 = \$230,000$
 $(\$300,000 - \$230,000)/\$230,000 = 30.4\%$
- $\$60,000 + \$40,000 + \$30,000 + \$20,000 = \$150,000$
 $(\$300,000 - \$150,000)/\$150,000 = 100\%$
- $\$60,000 + \$40,000 + \$30,000 = \$130,000$
 $(\$300,000 - \$130,000)/\$130,000 = 130.8\%$
- $\$60,000 + \$40,000 + \$30,000 + \$50,000 = \$180,000$
 $(\$300,000 - \$180,000)/\$180,000 = 66.7\%$

Objective: 5

156. Timothy Company has budgeted sales of \$780,000 with the following budgeted costs:

Direct materials	\$168,000
Direct manufacturing labor	132,000
Factory overhead	
Variable	96,000
Fixed	108,000
Selling and administrative expenses	
Variable	72,000
Fixed	100,000

Compute the average markup percentage for setting prices as a percentage of:

- Total manufacturing costs
- The variable cost of the product
- The full cost of the product
- Variable manufacturing costs

Answer:

- $\$168,000 + \$132,000 + \$96,000 + \$108,000 = \$504,000$
 $(\$780,000 - \$504,000)/\$504,000 = 54.8\%$
- $\$168,000 + \$132,000 + \$96,000 + \$72,000 = \$468,000$
 $(\$780,000 - \$468,000)/\$468,000 = 66.7\%$
- $\$168,000 + \$132,000 + \$96,000 + \$108,000 + \$72,000 + \$100,000 = \$676,000$
 $(\$780,000 - \$676,000)/\$676,000 = 15.4\%$
- $\$168,000 + \$132,000 + \$96,000 = \$396,000$
 $(\$780,000 - \$396,000)/\$396,000 = 97\%$

Objective: 5

157. Henderson Company is in the process of evaluating a new part using the following information.
- Part SLC2002 has one production run each month, each with \$16,000 in setup costs.
 - Part SLC2002 incurred \$40,000 in development costs and is expected to be produced over the next three years.
 - Direct costs of producing Part SLC2002 are \$56,000 per run of 24,000 parts each.
 - Indirect manufacturing costs charged to each run are \$88,000.
 - Destination charges for each run average \$18,000.
 - Part SLC2002 is selling for \$12.50 in the United States and \$25 in all other countries. Sales are one-third domestic and two-thirds exported.
 - Sales units equal production units each year.

Required:

- a. What are the estimated life-cycle revenues?
- b. What is the estimated life-cycle operating income for the first year?

Answer:

a. Domestic ($\$12.50 \times 12 \text{ months} \times 24,000 \times 3 \text{ yrs.} \times 1/3$)	\$ 3,600,000	
Export ($\$25 \times 12 \text{ months} \times 24,000 \times 3 \text{ yrs.} \times 2/3$)	<u>14,400,000</u>	
Estimated life-cycle revenues		<u>\$18,000,000</u>
b. Sales		
Domestic ($\$12.50 \times 12 \text{ months} \times 24,000 \times 1/3$)	\$1,200,000	
Export ($\$25 \times 12 \text{ months} \times 24,000 \times 2/3$)	<u>4,800,000</u>	
Total Sales		6,000,000
Costs:		
Development costs	\$ 40,000	
Setup costs ($12 \times \$16,000$)	192,000	
Direct manufacturing costs ($12 \times \$56,000$)	672,000	
Indirect manufacturing costs ($12 \times \$88,000$)	1,056,000	
Destination costs ($12 \times \$18,000$)	<u>216,000</u>	<u>2,176,000</u>
Estimated life-cycle operating income, first year		<u>\$3,824,000</u>

158. Stone and Bicker are starting a new business venture and are in the process of evaluating their product lines. Information for one new product, hand-made lamps, is as follows:
- Every six months a new lamp pattern will be put into production. Each new pattern will require \$11,200 in setup costs.
 - The lamp product line incurred \$48,000 in development costs and is expected to be produced over the next six years.
 - Direct costs of producing the lamps average \$144 each. Each lamp requires 12 labor-hours and 2 machine-hours.
 - Indirect manufacturing costs are estimated at \$160,000 per year.
 - Customer service expenses average \$16 per lamp.
 - Current sales are expected to be 2,000 units of each lamp pattern. Each lamp sells for \$224.
 - Sales units equal production units each year.

Required:

- a. What are the estimated life-cycle revenues?
- b. What is the estimated life-cycle operating income for the first year?

Answer:

a. Estimated life-cycle revenues:		
(2,000 x 2 patterns per year x \$224 per lamp)		\$ 896,000
		<u>x 6 years</u>
		<u>\$5,376,000</u>
b. Annual revenues (2,000 x \$224 x 2)		\$896,000
Setup costs (\$11,200 x 2)	\$ 22,400	
Development costs	48,000	
Direct manufacturing costs (2,000 x \$144 x 2)	576,000	
Indirect manufacturing costs	160,000	
Customer service costs (\$16 x 2,000 lamps x 2)	<u>64,000</u>	<u>870,400</u>
Estimated life-cycle operating income for the first year		<u>\$ 25,600</u>

CRITICAL THINKING

159. Claudia Geer, controller, discusses the pricing of a new product with the sales manager, James Nolan. What major influences must Claudia and James consider in pricing the new product? Discuss each briefly.

Answer:

The major influences are customers, competitors, and costs.

Customers: Managers must always examine pricing problems through the eyes of their customers. A price increase may cause customers to reject a company's product and choose a competing or substitute product.

Competitors: Competitors' reactions influence pricing decisions. At one extreme, a rival's prices and products may force a business to lower its prices to be competitive. At the other extreme, a business without a rival in a given situation can set higher prices. A business with knowledge of its rivals' technology, plant capacity, and operating policies is able to estimate its rivals' costs, which is valuable information in setting competitive prices.

Costs: Companies price products to exceed the costs of making them. The study of cost-behavior patterns gives insight into the income that results from different combinations of price and output quantities sold for a particular product.

160. In target costing, what are at least two techniques used to achieve target costing goals?

Answer:

In target costing, techniques used to achieve target-costing goals include value engineering, cross-functional teams, and supply-chain management.

161. What is the primary reason a firm would adopt target costing?

Answer:

The primary reason a firm would adopt target costing is to reduce costs. Its unique approach is to design costs out of products during the design stage in the product life cycle. Many firms are adopting this approach when they cannot reduce costs further using traditional costing methods, which focus on cost reductions in manufacturing.

162. Compare target costing and kaizen costing.

Answer:

Target costing focuses on reducing costs for products during the design stage. Kaizen costing focuses on reducing costs for products in the manufacturing stage.

Objectives: 3, 4

163. Ski Valet provides materials that let people teach themselves how to snow ski. It has six different skill-level programs. Each one includes visual and audio learning aids along with a workbook that can be submitted to the company for grading and evaluation purposes, if the person so desires.

The accounting system of Ski Valet is very traditional in its reporting functions with the calendar year being the company's fiscal year. It does include an abundance of information that can be used for various reporting purposes.

The company has found that any new idea soon runs its course with an effective life of about three years. Therefore, the company is always in the development stage of some new program. Program development requires experts in the area to provide the know-how of the item being developed and a development team that puts together the video, audio, and workbook materials. The actual costs of reproducing the packages are relatively inexpensive when compared to the development costs.

Required:

How might product life-cycle budgeting aid the company in improving its overall operations?

Answer:

Because the product life cycle for Ski Valet extends over several traditional accounting periods, it is critical for the company to consider a planning concept that evaluates each one of its products during its entire life cycle. Procedures that highlight an entire life cycle can include items for overall profitability, and which products might be repeated in a few years. With a large portion of their expenses in the development area, life-cycle budgeting can assist in predicting the sales needs for the entire life of a product.

It is probably more important to evaluate company performance on a product basis rather than year to year. Life-cycle budgeting would allow the company to compare products to each other rather than just comparing one year to the next.

164. What factors may influence the level of markups?

Answer:

Factors affecting the level of markups include the strength of demand, the elasticity of demand, and the intensity of competition. In addition, strategic reasons also may influence the level of markups. For instance, a firm may either choose a low markup to penetrate the market and win market share from established products of its competitors, or employ a high markup if it employs a skimming strategy for a market segment in which some customers are willing to pay higher prices for the privilege of owning the product.

Objectives: 5, 7

165. A hotel in Orlando, Florida, experiences peak periods and slower times. How should prices be adjusted during peak periods? During slow times? Why?

Answer:

During peak periods the hotel can justify increased prices because of full capacity conditions, whereas in slower periods when there is excess capacity, the hotel may want to lower prices to fill the excess capacity.

166. Clark Manufacturing offers two product lines, IN2 and EL5. The demand of the IN2 product line is inelastic, while the demand of the EL5 product line is very elastic. If Clark initiates a price increase for both product lines, how will customer demand change? How will the price increase affect operating profits?

Answer:

For the inelastic product line, when prices are increased demand will stay approximately the same and profits would be expected to increase. For the elastic product line, the increased price will result in decreased demand (i.e., lower sales volume). Whether a profit or a loss results from this change will depend on the amount of decreased demand and the amount of the increased contribution margin due to the increase in price.